# ETALE SPACES OVER Y GLUE ACROSS A MODIFICATION ISOMORPHIC OFF Z

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 996-1014 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 1016-1041, the complete author proof (24 body lines).

*Editorial note.* Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 23 of queue_spaces-pushouts.json (pb=1273, stmt 996-1014, proof 1016-1041).

*Editorial note.* spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 0GFY: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 888–905. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0GF6: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 1041–1041. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 03EB: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 1041–1041. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $S$ be a scheme. Let $f : Y' \to Y$ be a proper morphism of algebraic spaces over $S$. Let $i : Z \to Y$ be a closed immersion. Set $E = Z \times_Y Y'$. Picture*

$$\begin{array}{ccc} E & \xrightarrow{j} & Y' \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ Z & \xrightarrow{i} & Y \end{array}$$

*Date*: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*If $f$ is an isomorphism over $Y \setminus Z$, then the functor*

$$Y_{spaces,étale} \longrightarrow Y'_{spaces,étale} \times_{E_{spaces,étale}} Z_{spaces,étale}$$

*is an equivalence of categories.*

*Proof.* Let $(V' \to Y', W \to Z, \alpha)$ be an object of the right hand side. Recall that $V'$, resp. $W$ gives rise to a representable sheaf $\mathcal{G}' = h_{V'}$ in $Sh(Y'_{spaces,étale}) = Sh(Y'_{étale})$, resp. $\mathcal{G} = h_W$ in $Sh(Z_{spaces,étale}) = Sh(Z_{étale})$, see Properties of Spaces, Section 03EB. The isomorphism $\alpha : V' \times_{Y'} E \to W \times_Z E$ determines an isomorphism $j_{small}^{-1}\mathcal{G}' \to g_{small}^{-1}\mathcal{G}$ of sheaves on $E$. By Lemma 0GFY we obtain a unique sheaf $\mathcal{F}$ on $Y$ pulling pack to $\mathcal{G}'$ and $\mathcal{G}$ compatibly with the isomorphism. By Properties of Spaces, Lemma 0GF6 we obtain an étale morphism $V \to Y$ of algebraic spaces corresponding to $\mathcal{F}$; we omit the verification of the set theoretic condition[1]. The given isomorphism $\mathcal{G}' \to f_{small}^{-1}\mathcal{F}$ and $\mathcal{G} \to i_{small}^{-1}\mathcal{F}$ corresponds to isomorphisms $V' \to V \times_Y Y'$ and $W \to V \times_Y Z$ compatible with $\alpha$ as desired. □

[1]It follows from the fact that $\mathcal{G}$ and $\mathcal{G}'$ satisfies the corresponding condition.